

Network Security Milestone 1 Report

Constructing a Secure Network Infrastructure

May 7, 2026

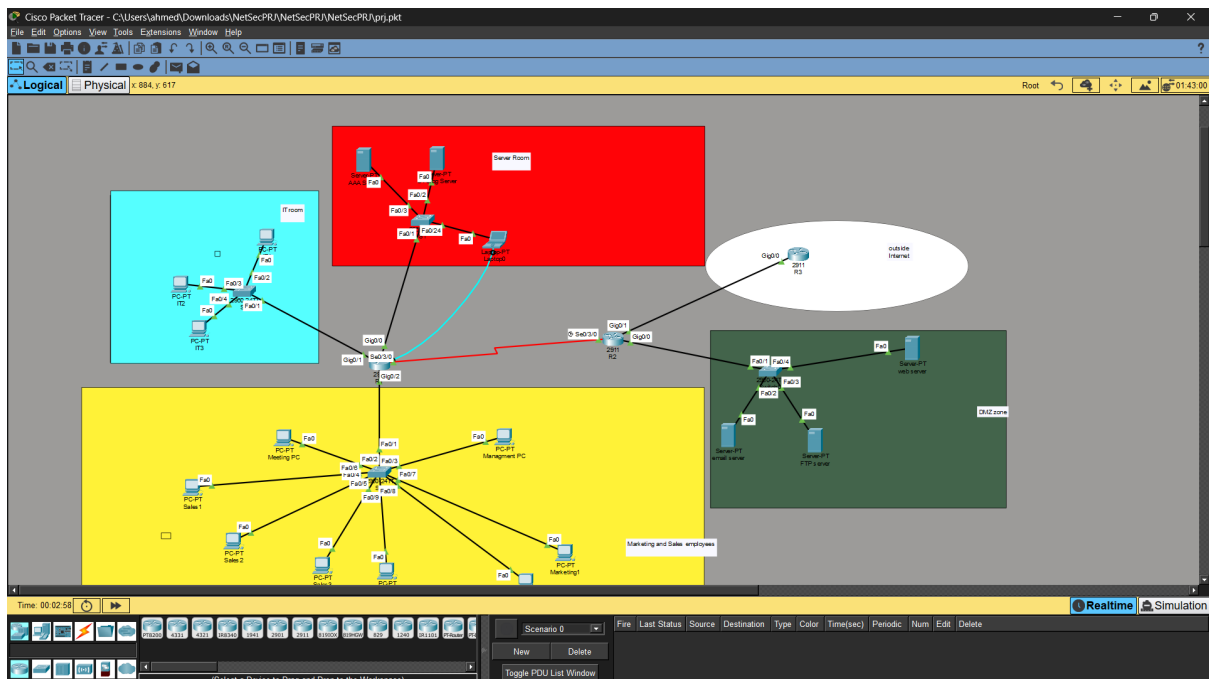
Contents

1	Introduction	3
2	Network Topology Overview	3
3	Configuration and Verification Steps	4
3.1	Basic Device Initialization	4
3.2	Hostname and Console Access	4
3.2.1	Configuration Commands (on R1)	4
3.2.2	Verification Commands	4
3.3	Interface Configuration and IP Addressing	4
3.3.1	Configuration Commands (on R1)	4
3.3.2	Verification Commands	5
3.4	Local User Accounts and Secure Shell (SSH) Configuration	5
3.4.1	Configuration Commands (on R1)	5
3.4.2	Verification Commands	5
3.5	OSPF Routing Protocol	5
3.5.1	Configuration Commands	6
3.5.2	Verification Commands	6
3.5.3	Evidence	6
3.6	TACACS+ AAA Authentication	6
3.6.1	Configuration Commands (on each router)	7
3.6.2	Verification Commands	7
3.6.3	Evidence	7
3.7	Syslog Logging	7
3.7.1	Configuration Commands (on each network device)	7
3.7.2	Verification Commands	7
3.7.3	Evidence	8
3.8	Save Configurations and Password Hashing	8
3.8.1	Configuration Commands	8
3.8.2	Verification Commands	8
3.8.3	Evidence	9
3.9	IOS IPS – Block Ping to Router R1	9
3.9.1	Configuration Commands (on R1)	9
3.9.2	Verification Commands	10
3.10	Access Control Lists (ACLs) for DMZ Security	10
3.10.1	Configuration Commands (on R1)	10
3.10.2	Verification Commands	11
3.10.3	Evidence	11
3.11	Brute-Force Attack Prevention	11
3.11.1	Configuration Commands (on each router)	11
3.11.2	Evidence	12

1 Introduction

This report documents the design and implementation of a secure, scalable network infrastructure for a small company using Cisco Packet Tracer. The network consists of internal departments (Server Room, Technical Support, Employees' Offices), a Demilitarized Zone (DMZ) for public-facing servers, and an external ISP connection. All devices are configured with OSPF dynamic routing, AAA authentication via TACACS+, centralised logging with Syslog, and various security controls including ACLs, IPS, and brute-force prevention.

2 Network Topology Overview



The network addressing scheme is summarised in Table 1.

Device/Subnet	IP Address / Network	Notes
Server Room	192.168.1.0/24	R1 interface: 192.168.1.1
Technical Support	192.168.2.0/24	R1 interface: 192.168.2.1
Employees' Offices	192.168.3.0/24	R1 interface: 192.168.3.1
DMZ	172.16.1.0/24	R2 interface: 172.16.1.1
R1–R2 Link	172.16.3.0/24	R1: 172.16.3.1, R2: 172.16.3.2
R2–R3 Link	172.16.2.0/24	R2: 172.16.2.1, R3: 172.16.2.2
Internet	209.165.200.0/24	R3: 209.165.200.1
AAA Server	192.168.1.10	TACACS+
Syslog Server	192.168.1.20	UDP 514
Web Server	172.16.1.30	HTTP
FTP Server	172.16.1.20	FTP
Email Server	172.16.1.10	SMTP/POP3

Table 1: IP addressing scheme

3 Configuration and Verification Steps

3.1 Basic Device Initialization

Before implementing advanced security features, all routers and switches must be configured with basic connectivity and secure management access. This section documents the initial setup performed on **Router R1** as a representative example. Identical steps were applied to Routers R2 and R3, with IP addresses adjusted according to Table 1.

3.2 Hostname and Console Access

3.2.1 Configuration Commands (on R1)

```
1 ! Enter global configuration mode
2 configure terminal
3
4 ! Set the device hostname
5 hostname R1
6
7 ! Configure console line
8 line console 0
9 login local
10 exit
```

3.2.2 Verification Commands

```
1 ! Check the running configuration for console settings
2 show running-config | section line con
```

3.3 Interface Configuration and IP Addressing

3.3.1 Configuration Commands (on R1)

```
1 interface gigabitEthernet 0/0
2 description Link to Server Room Switch S1
3 ip address 192.168.1.1 255.255.255.0
4 no shutdown
5 exit
6
7 interface gigabitEthernet 0/1
8 description Link to Tech Support Switch S2
9 ip address 192.168.2.1 255.255.255.0
10 no shutdown
11 exit
12
13 interface gigabitEthernet 0/2
14 description Link to Employees Switch S3
15 ip address 192.168.3.1 255.255.255.0
16 no shutdown
17 exit
18
19 interface serial 0/3/0
20 description Link to Edge Router R2
21 ip address 172.16.0.1 255.255.255.252
```

```
22 no shutdown
23 exit
```

3.3.2 Verification Commands

```
1 show ip interface brief
2 show interfaces status
3 ping 192.168.1.1 ! verify interface is up
```

3.4 Local User Accounts and Secure Shell (SSH) Configuration

The project specification requires a local console user Admin1 and a remote SSH user SSHAdmin1. All routers must support SSH version 2 for secure remote management.

3.4.1 Configuration Commands (on R1)

```
1 ! Create local user accounts with hashed passwords
2 username Admin1 secret admin1Pa55
3 username SSHAdmin1 secret SSHadmin1Pa55
4
5 ! Configure domain name (required for RSA key generation)
6 ip domain name company.local
7
8 ! Generate RSA key pair for SSH
9 crypto key generate rsa modulus 1024
10
11 ! Enable SSH version 2
12 ip ssh version 2
13
14 ! Secure VTY lines for SSH access only
15 line vty 0 4
16   transport input ssh
17   login local
18
19 exit
20
21 ! Enable password encryption service globally
22 service password-encryption
```

3.4.2 Verification Commands

```
1 ! Confirm SSH is enabled and key is generated
2 show ip ssh
3
4 ! Verify local user database
5 show running-config | include username
6
7 ! Attempt SSH connection from a PC to R1
8 ! (from PC command prompt: ssh -l SSHAdmin1 192.168.1.1)
```

3.5 OSPF Routing Protocol

Objective: Configure OSPF on all routers (R1, R2, R3) to ensure full connectivity.

3.5.1 Configuration Commands

```

1 router ospf 1
2   network 192.168.1.0 0.0.0.255 area 0
3   network 192.168.2.0 0.0.0.255 area 0
4   network 192.168.3.0 0.0.0.255 area 0
5   network 172.16.3.0 0.0.0.255 area 0

```

Listing 1: OSPF configuration on R1

```

1 router ospf 1
2   network 172.16.1.0 0.0.0.255 area 0
3   network 172.16.2.0 0.0.0.255 area 0
4   network 172.16.3.0 0.0.0.255 area 0

```

Listing 2: OSPF configuration on R2

```

1 router ospf 1
2   network 172.16.2.0 0.0.0.255 area 0
3   network 209.165.200.0 0.0.0.255 area 0

```

Listing 3: OSPF configuration on R3

3.5.2 Verification Commands

```

1 show ip ospf neighbor
2 show ip route ospf
3 ping 172.16.1.10 source 192.168.3.10 ! from Sales PC to Web Server

```

Listing 4: Verify OSPF neighbors and routing table

3.5.3 Evidence

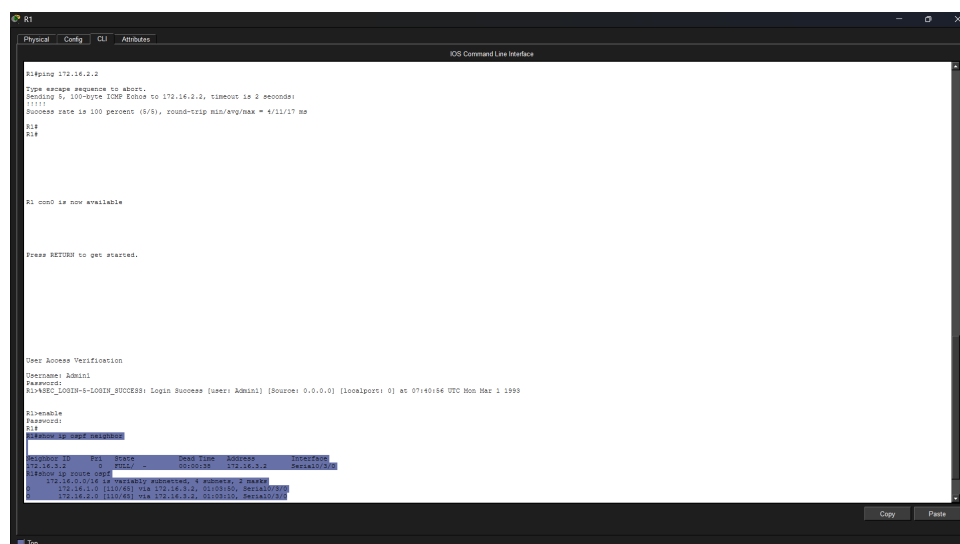


Figure 1: Output of show ip ospf neighbor on R1 showing FULL state with R2

3.6 TACACS+ AAA Authentication

Objective: Centralise authentication of all routers using a TACACS+ server.

3.6.1 Configuration Commands (on each router)

```
1 aaa new-model
2 tacacs-server host 192.168.1.10 key cisco123
3 aaa authentication login default group tacacs+ local
4 aaa authorization exec default group tacacs+ local
5 aaa accounting exec default start-stop group tacacs+
```

3.6.2 Verification Commands

```
1 test aaa group tacacs+ SSHAdmin1 SSHAdmin1Pa55 legacy
2 show aaa sessions
```

Attempt SSH login to R1 using the remote credentials.

3.6.3 Evidence

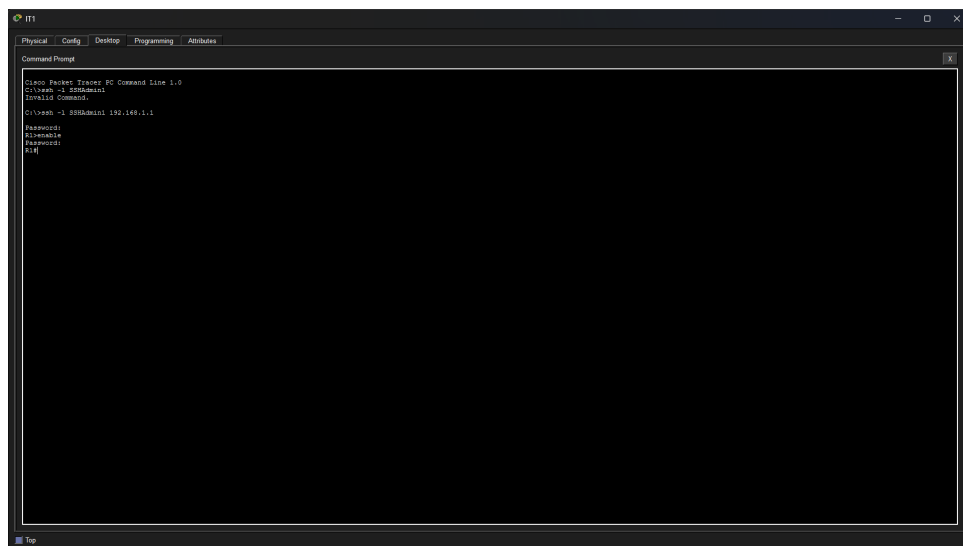


Figure 2: Successful SSH login to R1 authenticated via TACACS+

3.7 Syslog Logging

Objective: Forward all system logs to a central Syslog server.

3.7.1 Configuration Commands (on each network device)

```
1 logging on
2 logging host 192.168.1.20
3 logging trap debugging
4 service timestamps log datetime msec
```

3.7.2 Verification Commands

```
1 show logging
2 ! On Syslog server: observe incoming logs in the Services tab
```

3.7.3 Evidence

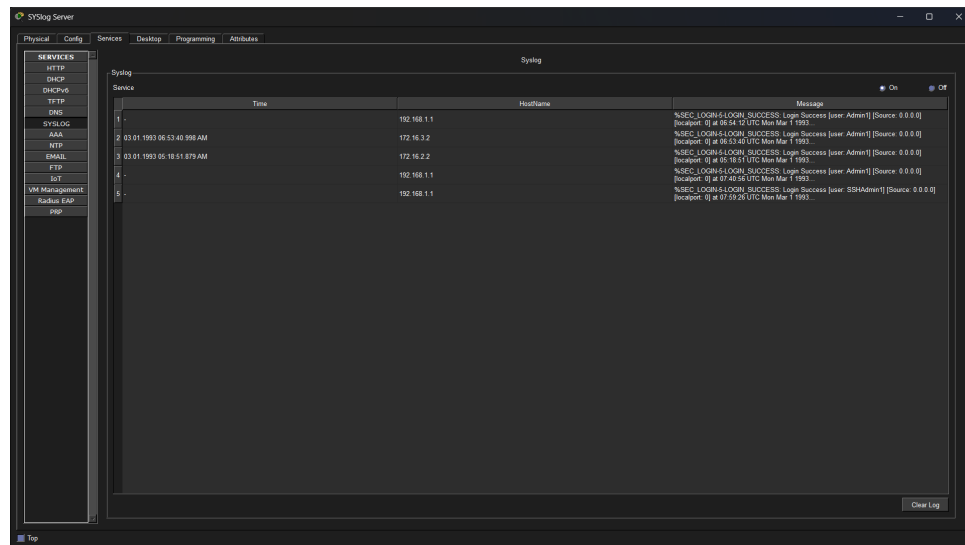


Figure 3: Syslog server displaying log messages from R1

3.8 Save Configurations and Password Hashing

Objective: Save running configuration to NVRAM and ensure all passwords are hashed.

3.8.1 Configuration Commands

```
1 enable secret class123
2 service password-encryption
3 copy running-config startup-config
```

3.8.2 Verification Commands

```
1 show startup-config | include enable secret
2 show startup-config | include password
```


3.8.3 Evidence

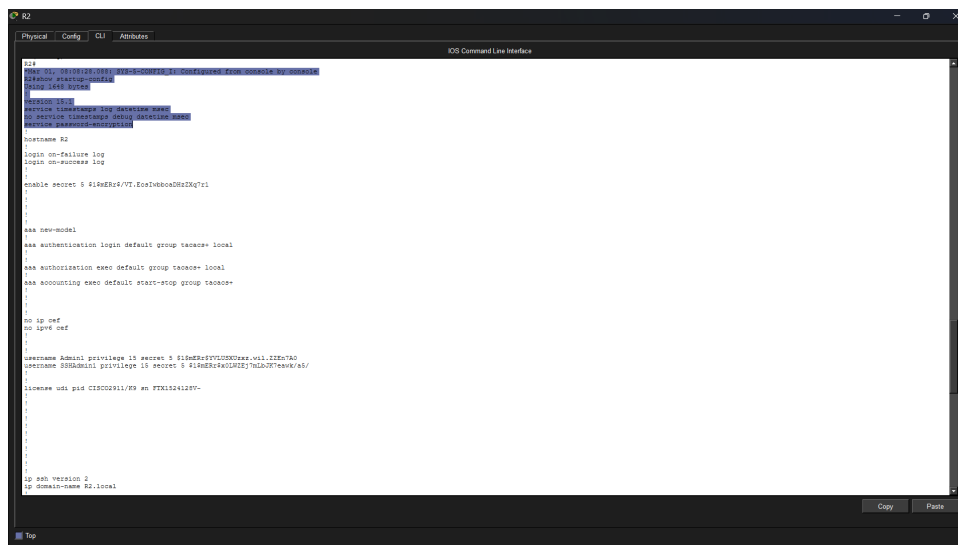


Figure 4: Output showing hashed enable secret (type 5) and encrypted line passwords (type 7)

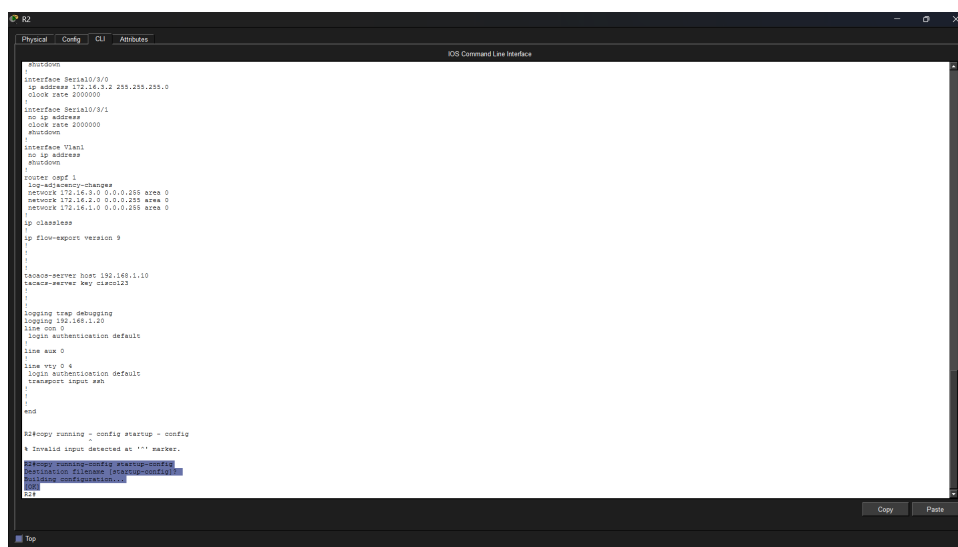


Figure 5: Output showing the copying of the running config into startup config (type 7)

3.9 IOS IPS – Block Ping to Router R1

Objective: Prevent any employee from pinging the Server Room router (R1) while allowing all other traffic (SSH, OSPF, etc.) to pass normally.

Note: Due to Cisco Packet Tracer’s IPS simulation behavior, the basic signature set remains active, and only signature 2004 (ICMP Echo Request) is modified to deny packets.

3.9.1 Configuration Commands (on R1)

```
1 ! Create IPS rule
2 ip ips name BLOCKING_PING
3
4 ! Keep basic signature set active (required in PT)
5 ip ips signature-category
6   category all
7   retired false
8   exit
9 exit
10
11 ! Modify signature 2004 to deny ICMP Echo Requests
12 ip ips signature-definition
13   signature 2004 0
14   status
15     retired false
16     enabled true
17   exit
18   engine
19     event-action produce-alert
20     event-action deny-packet-inline
21   exit
22 exit
23 exit
24
25 ! Apply IPS rule inbound on internal-facing interfaces
26 interface gigabitEthernet 0/0
27   ip ips BLOCKING_PING out
28 exit
```

3.9.2 Verification Commands

```
1 show ip ips interfaces
2 show ip ips statistics
```

3.10 Access Control Lists (ACLs) for DMZ Security

Objective: Restrict access to DMZ servers according to company policy:

- Permit Sales and Marketing employees to access the Web Server.
- Deny the Meeting Room PC from accessing the FTP Server.

Design Note: The extended ACL is applied in on Router R1's internal interface (GigabitEthernet 0/2) to filter traffic as close to the source as possible. A final **permit ip any any** rule ensures that other necessary traffic (internal routing, SSH, etc.) is not blocked.

3.10.1 Configuration Commands (on R1)

```
1 ! Create extended ACL 101 with proper ordering (specific rules first)
2 access-list 101 remark == DMZ Traffic Control ==
3
4 ! Deny Meeting Room PC FTP access (both control and data ports)
5 access-list 101 deny tcp host 192.168.3.30 host 172.16.1.20 eq 21
```

```

6 access-list 101 deny tcp host 192.168.3.30 host 172.16.1.20 eq 20
7
8 ! Permit Sales/Marketing subnet to Web Server (HTTP/HTTPS)
9 access-list 101 permit tcp 192.168.3.0 0.0.0.255 host 172.16.1.30 eq 80
10 access-list 101 permit tcp 192.168.3.0 0.0.0.255 host 172.16.1.30 eq
    443
11
12 ! Permit all other internal IP traffic to DMZ (Email, etc.)
13 access-list 101 permit ip 192.168.0.0 0.0.255.255 172.16.1.0 0.0.0.255
14
15 ! Allow all remaining traffic (internal communication, SSH, OSPF, etc.)
16 access-list 101 permit ip any any
17
18 ! Apply ACL inbound on the Employees subnet interface
19 interface gigabitEthernet 0/2
20 ip access-group 101 in
21 exit

```

3.10.2 Verification Commands

```

1 show access-lists 101
2 show running-config interface gigabitEthernet 0/2

```

3.10.3 Evidence

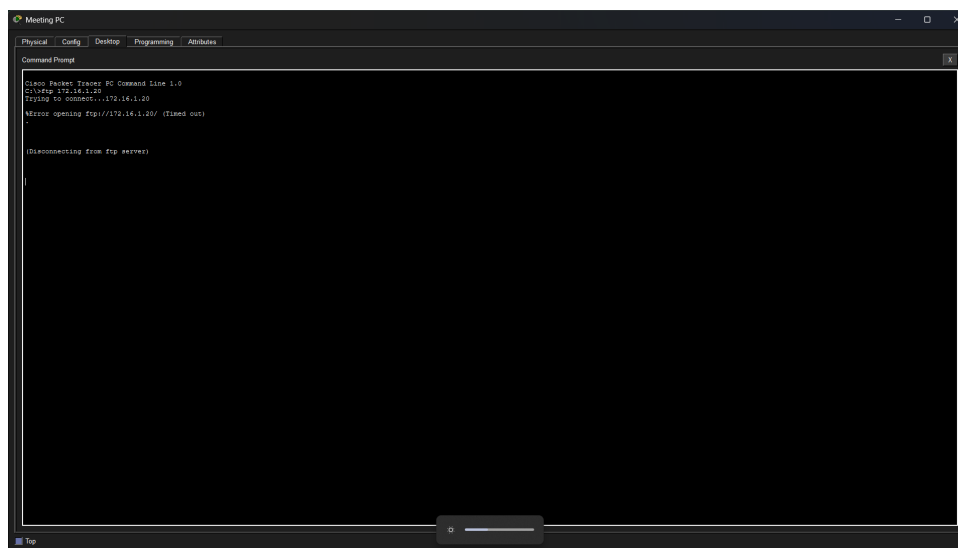


Figure 6: FTP connection fails from Meeting Room PC; ACL counters show deny matches

3.11 Brute-Force Attack Prevention

Objective: Block any user for 5 minutes after 3 failed login attempts within 60 seconds.

3.11.1 Configuration Commands (on each router)

```

1 ! Enable login blocking with specified parameters
2 login block-for 300 attempts 3 within 60

```

3.11.2 Evidence

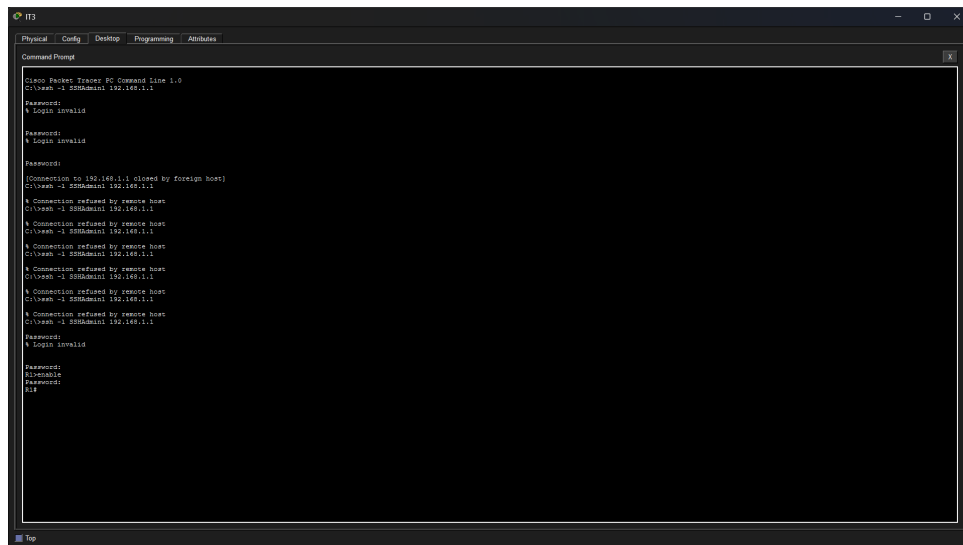


Figure 7: `show login` displays the blocked IP and remaining block time after 3 failed attempts